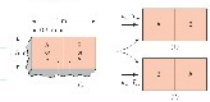


7.17, 7.38, 8.25, 8.66 CH7: External Flow CH8: Internal Flow

7.17 Flat Plate in Parallel Flow

7.17 The two cases in a convection problem of case 1 and case 2 are shown. In case 1, the surface is initially smooth and the flow is laminar. In case 2, the surface is initially rough and the flow is turbulent. The roughness is shown in the diagram. The roughness is shown in the diagram. The roughness is shown in the diagram.



- min total convection heat transfer to the surface?
- rate of convection heat transfer to the entire surface

case 1, flow is initially smooth/laminar, then becomes turbulent
 case 2, flow is initially turbulent, then becomes smoother
 Turbulent flow results in higher conduction so case 1 is better as it minimizes total convection heat transfer.

case 1: $T_f = \frac{10 + 35}{2} = 22.5^\circ\text{C} \rightarrow 295.6^\circ\text{K}$

Appendix A-4: $T_f \approx 300^\circ\text{K}$ $\nu = 15.89 \times 10^{-6} \text{ m}^2/\text{s}$, $K = 263 \times 10^{-3} \text{ W/m}\cdot\text{K}$, $Pr = .707$

$Re = \frac{U_\infty L}{\nu} = \frac{(25 \text{ m/s})(1 \text{ m})}{(15.89 \times 10^{-6} \text{ m}^2/\text{s})} = 1.6 \times 10^6$

$Pr = .707 \rightarrow \text{eqn 7.46} \rightarrow$

$Nu_L = (0.0308 Re_L^{1/4} - A) Pr^{1/4} \rightarrow$

$Nu_L = 0.0308 ((1.6 \times 10^6)^{1/4} - 871) (.707)^{1/4} = 1745$

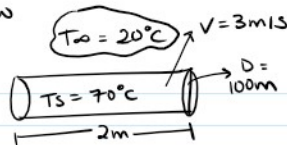
$\bar{h}_L = \frac{Nu_L \cdot K}{L} = \frac{1745 (263 \times 10^{-3})}{1} = 45.9 \frac{\text{W}}{\text{m}^2 \cdot \text{K}}$

$q_1 = \bar{h}_L A (T_s - T_\infty) = (45.9) (.5 \text{ m}^2) (10 - 35) = -25 \text{ W}$

$q_1 = -25 \text{ W}$

7.38 Cylinder in Cross Flow

7.38 Determine the convection heat transfer coefficient, thermal resistance for convection, and the convection heat transfer rate that are associated with air at atmospheric pressure in cross flow over a cylinder of diameter $D = 100 \text{ mm}$ and length $L = 2 \text{ m}$. The cylinder temperature is $T_s = 70^\circ\text{C}$ while the air velocity and temperature are $V = 3 \text{ m/s}$ and $T_\infty = 20^\circ\text{C}$, respectively. Plot the convection heat transfer coefficient and the heat transfer rate from the cylinder over the range $0.05 \leq D \leq 0.5 \text{ m}$.



$T_f = \frac{70 + 20}{2} = 45^\circ\text{C}$

Thermophysical Properties @ 45°C

$\rho = 1.109 \text{ kg/m}^3$

$K = .02699 \text{ W/m}\cdot\text{K}$

$\mu = 1.941 \times 10^{-5} \text{ kg/m}\cdot\text{s}$

$\nu = 1.750 \times 10^{-5} \text{ m}^2/\text{s}$

$Pr = .7241$

$Re = \frac{VD}{\nu} = \frac{(3 \text{ m/s})(0.1 \text{ m})}{1.750 \times 10^{-5} \text{ m}^2/\text{s}} = 17142$

$Re = 17142$

$\bar{Nu}_D \equiv \frac{hD}{K} = C Re_D^m Pr^{1/4}$

$\bar{Nu}_D = (193) (17142)^{.618} (.7241)^{1/4} = 72$

$\bar{h} = \frac{K \cdot \bar{Nu}_D}{D} \rightarrow \frac{(.02699 \text{ W/m}\cdot\text{K}) (72)}{(.1 \text{ m})} = 19.3 \text{ W/m}^2 \cdot \text{K}$

$\bar{h} = 19.3 \text{ W/m}^2 \cdot \text{K}$

$A = \pi D L = \pi (.1)(2) = .6283 \text{ m}^2$

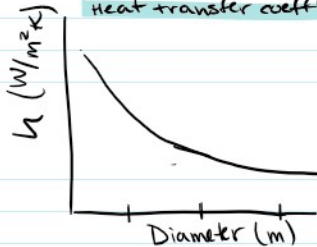
$R_{th} = \frac{1}{hA} = \frac{1}{(19.3)(.6283)} \Rightarrow R_{th} = .082 \frac{\text{K}}{\text{W}}$

$$R_{th} = \frac{1}{hA} = \frac{1}{(19.3)(.6283m^2)} \Rightarrow R_{th} = .082 \frac{K}{W}$$

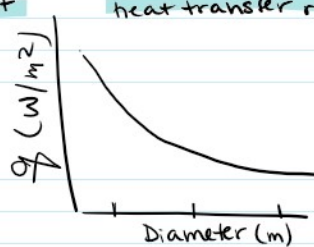
$$q = \frac{\Delta T}{R_{th}} \frac{1}{A} = \frac{(70-20)}{(.082)} \frac{1}{.6283m^2}$$

$$q = 970 \text{ W/m}^2$$

Heat transfer coefficient



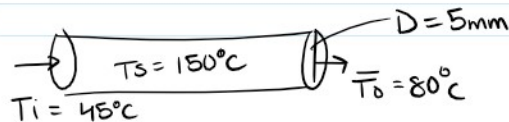
Heat transfer rate



"general diagram"

8.25 Heat Transfer Correlations: Circular Tubes

8.25 Engine oil flows at a rate of 1 kg/s through a 5-mm-diameter straight tube. The oil has an inlet temperature of 45°C and it is desired to heat the oil to a mean temperature of 80°C at the exit of the tube. The surface of the tube is maintained at 150°C. Determine the required length of the tube. Hint: Calculate the Reynolds numbers at the entrance and exit of the tube before proceeding with your analysis.



Required length of tube?

@ inlet

$$273 + 45 = 318^\circ K \approx 320^\circ K$$

$$\mu = 14.1 \text{ N}\cdot\text{s/m}^2$$

$$Re = \frac{4m}{\pi D \mu} = \frac{4(1)}{\pi (.005m)(14.1 \times 10^2)} = 1806 \text{ Laminar}$$

@ outlet

$$273 + 80 = 353^\circ K \approx 350^\circ K$$

$$\mu = 3.56 \text{ N}\cdot\text{s/m}^2$$

$$Re = \frac{4m}{\pi D \mu} = \frac{4(1)}{\pi (.005m)(3.56 \times 10^2)} = 7153 \text{ Turbulent}$$

Laminar (Region 1)

$$T_m = 320^\circ K \rightarrow$$

$$\left. \begin{array}{l} C_p = 1.993 \text{ kJ/kg}\cdot\text{K} \\ \mu_1 = 14.1 \times 10^2 \text{ N}\cdot\text{s/m}^2 \\ K_1 = 143 \times 10^3 \text{ W/m}\cdot\text{K} \\ Pr_1 = 1965 \end{array} \right\} Re_1 = \frac{4m}{\pi D \mu_1} = \frac{4(1)}{\pi (.005)(14.1 \times 10^2)} = 1806 \checkmark$$

$$X_{fd,h} = .05 Re D = (.05)(1806)(.005) = .45m$$

$$X_{fd,t} = X_{fd,h} \cdot Pr = (.45m)(1965) = 887$$

$$\frac{T_s - T_m}{T_s - T_i} = e \left[\frac{-\pi D L_1 \bar{h}_1}{m C_p} \right] \rightarrow h_1 L_1 = \frac{m C_p}{\pi D} \ln \left[\frac{T_s - T_m}{T_s - T_i} \right] = \frac{1 \times 1999}{\pi (.005)} \ln \left(\frac{150 - 52}{150 - 45} \right) = 8780$$

$$\text{if } Nu = 3.66, Nu = \frac{3.66 + .0668(D/L_1) Re_1 Pr_1}{1 + .04(D/L_1) Re_1 Pr_1^{2/3}} \neq \frac{\bar{h}_1 D}{K_1} \left\{ h_1 = \frac{3.66 K_1}{D} = 105 \right.$$

$$Nu_1 = \bar{h}_1 D \rightarrow 16.9 K_1 = h_1 = (16.9)(.143) = 483 \text{ W/m}^2\cdot\text{K}$$

show $Nu \approx 17$

$$Nu_1 = \frac{h_1 D}{K_1} \rightarrow \frac{16.9 K_1}{D} = h_1 = \frac{(16.9)(.143)}{(.005)} = 483 \text{ W/m}^2 \cdot \text{K}$$

$$483 \times L_1 = 8780 \rightarrow \boxed{L_1 = 18.2 \text{ m}}$$

TURBULENT (Region 2)

TABLE A5 $\bar{T}_{m2} = \frac{52+80}{2} \approx 340^\circ \text{K} \rightarrow$

$$\left. \begin{aligned} c_{p2} &= 2.076 \text{ kJ/kg} \cdot \text{K} \\ \mu_2 &= 5.81 \times 10^{-2} \text{ N} \cdot \text{s/m}^2 \\ k_2 &= 139 \times 10^3 \text{ W/m} \cdot \text{K} \\ Pr_2 &= 793 \end{aligned} \right\}$$

$$Re_2 = \frac{4m}{\pi D \mu} = \frac{4(1)}{\pi (.005)(5.81 \times 10^{-2})} = 4795 \checkmark$$

— friction factor —
 $f = (1.790 \ln(4795) - 1.64)^{-2} \Rightarrow f = .039$

$$Nu_2 = \frac{(f/8)(Re_2 - 1000)Pr_2}{1 + 12.7(f/8)^{1/2}(Pr_2^{1/3} - 1)} = \frac{(.039/8)(4795 - 1000)793}{1 + 12.7(.039/8)^{1/2}(793^{1/3} - 1)} = 192$$

eqn 8.62)

$$\therefore \bar{h}_2 = \frac{Nu_2 K_2}{D} = \frac{(192)(.139)}{(.005)} = 5360 \text{ W/m}^2 \cdot \text{K}$$

$$h_2 L_2 = \frac{m c_{p2}}{\pi D} \ln \left[\frac{T_s - T_m}{T_s - T_c} \right] \rightarrow L_2 = \frac{m c_{p2}}{\pi D h_2} \ln \left[\frac{T_s - T_0}{T_s - T_m} \right] = \frac{(1)(2076)}{(\pi)(.005)(5360)} \ln \left[\frac{180 - 80}{180 - 52} \right] = 8.3 \text{ m}$$

$$\therefore L = L_1 + L_2 = 18.2 + 8.3 \Rightarrow \boxed{L = 26.5 \text{ m}}$$

8.66 Noncircular Ducts

8.66 Consider laminar, fully developed flow in a channel of constant surface temperature T_s . For a given mass flow rate and channel length, determine which rectangular channel, $b/a = 1.0, 1.43$, or 2.0 , will provide the highest heat transfer rate. Is this heat transfer rate greater than, equal to, or less than the heat transfer rate associated with a circular tube?

$T_s \checkmark$
 $m \checkmark$
 $L \checkmark$
 greatest heat transfer rate?
 ... circular tube?



uniform T_s
 $\frac{b}{a} = 1.0, Nu = 2.98, \frac{\bar{h} D_h}{K} = 2.98$

$$\boxed{h = \frac{2.98 K}{a}}$$

$$D_h = a$$

eqn 8.66
 $D_h = \frac{4A}{P}$

$\frac{b}{a} = 1.43, Nu = 3.08, \frac{\bar{h} D_h}{K} = 3.08$

$$h = \frac{3.08 K}{D_h}$$

$$\boxed{h = \frac{2.704 K}{a}}$$

$$D_h = \frac{4(a \times 1.43a)}{2(1 + 1.43)a} = 1.176a$$

$\frac{b}{a} = 2.0, Nu = 3.39$

$$h = \frac{3.39 K}{D_h}$$

$$\boxed{h = 2.543 \frac{K}{a}}$$

$$D_h = \frac{4(a \times 2a)}{2(a + 2a)} = \frac{4}{3}a$$

$\therefore \frac{b}{a} = 1$ will provide the greatest heat transfer!

circular tube,
 $Nu = 3.66$

16

$$h = 366 \frac{K}{a}$$

heat transfer rate
is greater in circular tube!